Question 1

Which of the following is true

A [1,2] ⊆(1, 3)

B [1,2] ⊆[1,3)

C [1,2] ⊆ {1,3}

D {1,3} ⊆ [1,2]

Answer - B

Question 4

8 Points

We know that if a knowledge base KB has two components A and B then to check if C is a logical consequence from KB we assemble the formula "If (A and B) then C" and then based on the classification of this formula determine if there is a logical consequence of C from KB. Determine which one of the following cases represents logical consequence between the knowledge base KB and a statement C.

A $KB = \{p \Rightarrow q, q \wedge \neg p\}, \quad C = p$

B $KB = \{\neg p \Rightarrow q, \neg p\}, C = p \wedge q$

C $KB = \{p \Rightarrow q, q \Rightarrow r\}, \quad C = p \Rightarrow r$

D $KB = \{p \Rightarrow \neg q, \neg q\}, \quad C = p \wedge q$

Answer - C

Question 5

4 Points

A graph G is defined as follows: V = {a,b,c,d,e} and E = {{a,b}, {b,c}, {c,d}, {e,e}}. Which statement below is true in relation to this graph

(A) G is acyclic

(B) G is disconnected

(C) G has five edges

(D) G has four nodes

Answer - B

Question 6

4 Points

How many times in the final column of the truth table for the following formula $(p \vee \neg p) \Rightarrow (q \wedge \neg q)$ the value "FALSE" would appear?

(A) 8

(B) 4

(C) 0

(D) 2

Answer - B

Question 7

3 Points

Which statement below is true in relation to graph G

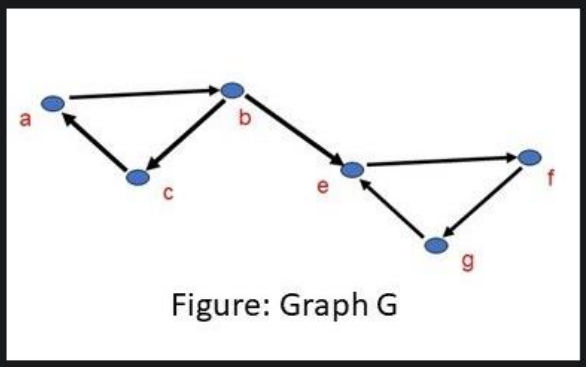


Figure: Graph G

A This graph has 7 nodes

B This graph is a tree with root node "a"

C This graph is a tree with root node "b"

D This is a directed cyclic graph

Answer - D

Question 8

4 Points

The only values for the variables p and q which make the formula

$\neg (p \Rightarrow q)$

true are

A p is TRUE, q is FALSE

B p is FALSE, q is TRUE

C p is FALSE, q is FALSE

D p is TRUE, q is TRUE

Answer - A

Question 9

4 Points

Let A = {p, q} and B = {0,1}
Which of the following relations is a function with domain A and co-domain B?

A {(p,0), (q,1), (q,0)}

B {(p,1),(q,0)}

C {(p,q), (0,1)}

D {(p,1),(q,0), (p,0)}

Answer - B

For a formula

$(\neg(p \Rightarrow \neg q) \vee p) \Rightarrow p$

We prioritise the Booleans:

3	2	1	4	5
$(\neg(p$	$\Rightarrow \neg q)$	$\vee p)$	$\Rightarrow$	p

One of the rows of an attempt below to build a truth table for this formula has a mistake. Your task is to identify which row it is.

			1		2	3		4		5	
	p	q	$\neg q$		$p \Rightarrow \neg q$	$\neg(p \Rightarrow \neg q)$		$\neg(p \Rightarrow \neg q) \vee p$		$(\neg(p \Rightarrow \neg q) \vee p) \Rightarrow p$	
1	T	T	F		F	T		T		T	
2	T	F	T		T	F		T		T	
3	F	T	F		T	F		F		T	
4	F	F	T		T	F		F		F	

A Row 1

B Row 3

C Row 2

D Row 4

Answer - D

Question 11

4 Points

How many rows would be in the truth table for the formula?

$\neg(\neg p \Rightarrow (q \wedge \neg q)) \vee (\neg r \Rightarrow (s \vee \neg s))$

A) 6

B) 8

C) 4

D) 16

Answer - D

Question 12

4 Points

Which set below is defined by the following notation, where Z is a set of integers:

$\{x \in Z: x^3 = -8\}$

A) {-2}

B) {2}

C) {}

D) {-2,2}

Answer – A

Question 13

6 Points

Let A be the interval [4,25). Which of the following is the set satisfying the condition $\{x \in A : x^2 \in A\}$

(A) {4, 5}

(B) [4, 5)

(C) (2,5)

(D) {}

Answer - B

Question 14

5 Points

What is the result of the following operation:
$(1,10) \cap \{10\}$

(A) (1,10)

(B) (1,10]

(C) {}

(D) {1,10}

Answer – C

Question 15

6 Points

Consider the following formula

$\neg\neg((p \Rightarrow q) \vee (q \Rightarrow p))$

What is the correct answer to the question: which input values for the variables *p* and *q* result in the value "FALSE" in the final column of the truth table.

A. None of the input values for the variables *p* and *q* result in the value "FALSE"

B. Only when *p* is FALSE and *q* is FALSE

C. All of the input values for the variables *p* and *q* result in the value "FALSE"

D. Only when *p* is TRUE and *q* is FALSE

Answer - A

Question 16

4 Points

Which set builder notation below defines the interval (1,100]

A. $\{x \in R : 1 \leq x \text{ and } x \leq 100\}$

B. $\{x \in R : 1 < x \text{ and } x < 100\}$

C. $\{x \in R : 1 \leq x \text{ and } x < 100\}$

D. $\{x \in R : 1 < x \ \& \text{ and } x \leq 100\}$

Answer - D

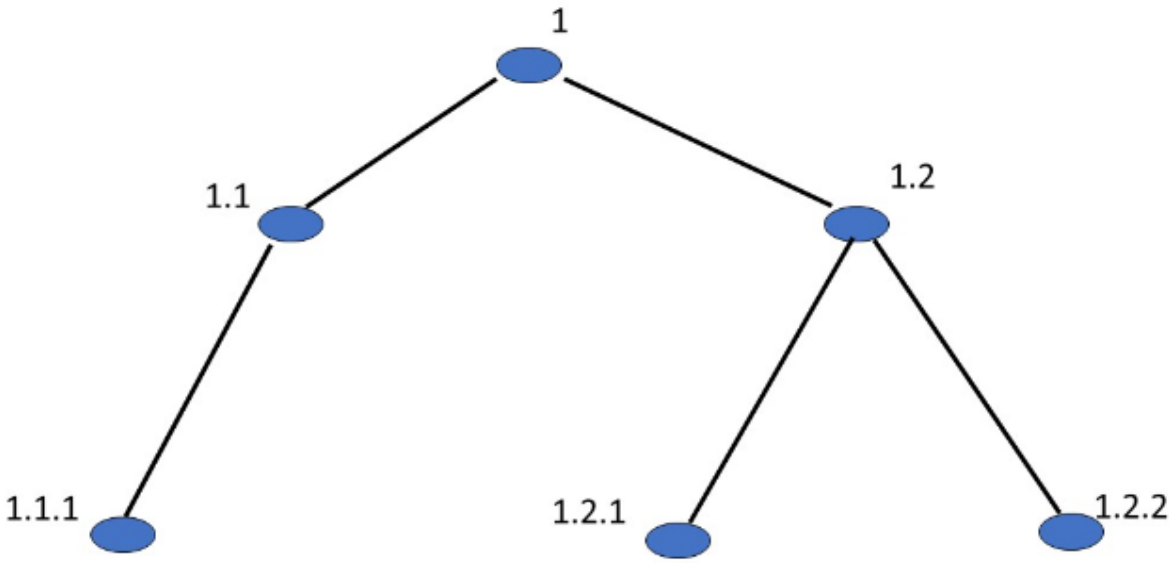


Figure: Graph T

Question 17 — 4 Points

Which statement below is true in relation to this graph T

A. This is not a binary tree because node 1.1 has only one child

B. This is a binary tree with root 1

C. Nodes 1.1.1 and 1.2.2 are disconnected

D. This is a disconnected graph

Answer – B

Question 18

4 Points

You are given the relation R = {(a,0), (b, 1), (c,1)} as a subset of AxB. In which situation this relation is a function with the domain A and co-domain B?

A. A ={a, b, c}, B = {}

B. A ={a, b, c}, B = {0}

C. A ={a, b, c}, B = {1,0}

D. A ={a, b, c, d}, B = {1,0}

Answer - C

Which sets V and E describe the following graph

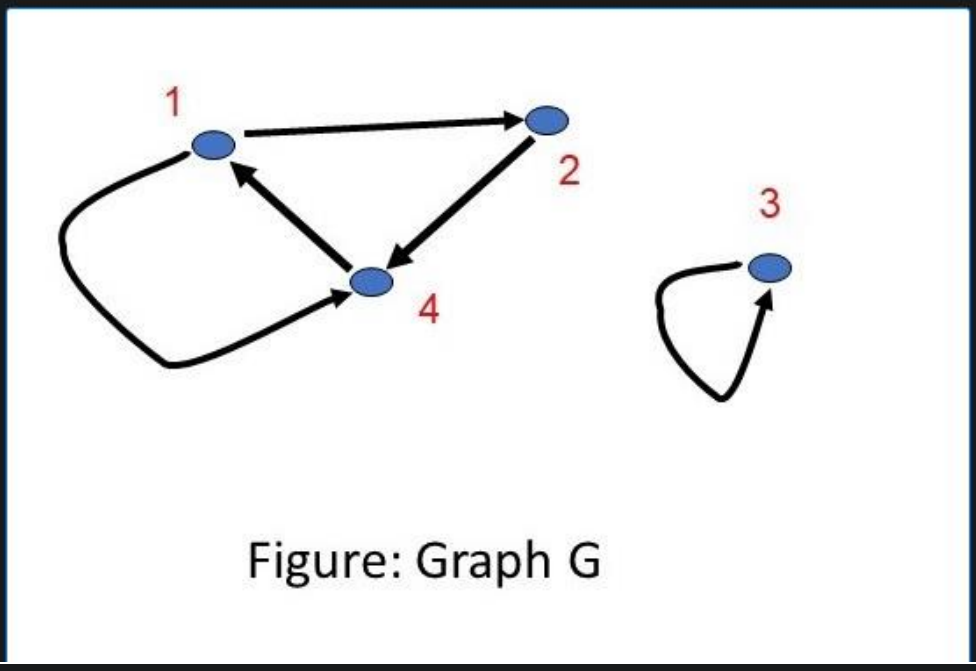


Figure: Graph G

A
V={1,2,3,4}
E={{1,2}, {2,4}, {4,1},{1,4}}

B
V={1,2,3,4}
E={(1,2), (2,4), (1,4),(3,3)}

C
•V={1,2,4}
•E={(1,2), (2,4), (4,1), (1,4)}

D
V={1,2,3,4}
E={(1,2), (2,4), (4,1),(1,4),(3,3)}

Answer – D

Question 20

7 Points

Let A={a,b,c,d, e} and B = {d,e,f,g,h,i}.
The cardinality of the set |BxA \ AxB| is

A 0

B 30

C 4

D 26

Answer - B

Which operation is reflected in the following Venn diagram

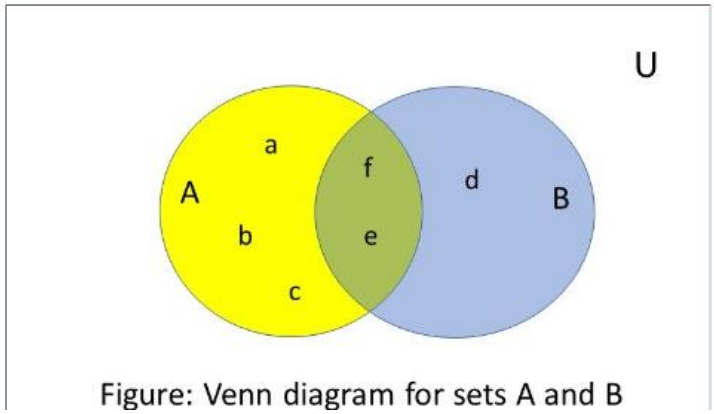


Figure: Venn diagram for sets A and B

A $(A-B) \cap (B-A) = \{\}$

B $(A \cap B) \cup \{d\} = A$

C $(B-A) \cup \{a,b,c\} = B$

D $A-(A \cap B) = \{e, f\}$

Answer - A